

VALIDATION OF RADAR-DEFINED HYDROMETEOR CLASSIFICATION VIA THE RADAR COLUMNS AND AIRCRAFT TIMESERIES (RADCAT) PRODUCT

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MOTIVATIONS

- The ability to compare cloud microphysics aircraft data with ground radar data to further our understanding of severe storm dynamics.
- Previous work by (Murphy et al. 2020) comparing these datasets but no open-source method is available.
- With this new synced dataset and CSU Radar Tools hydrometeor classification, a better understanding of biases that might be present within the classification algorithm could be discovered.



Figure 1: University of North Dakota Citation II microphysics aircraft during MC3E field campaign (Collis 2011).

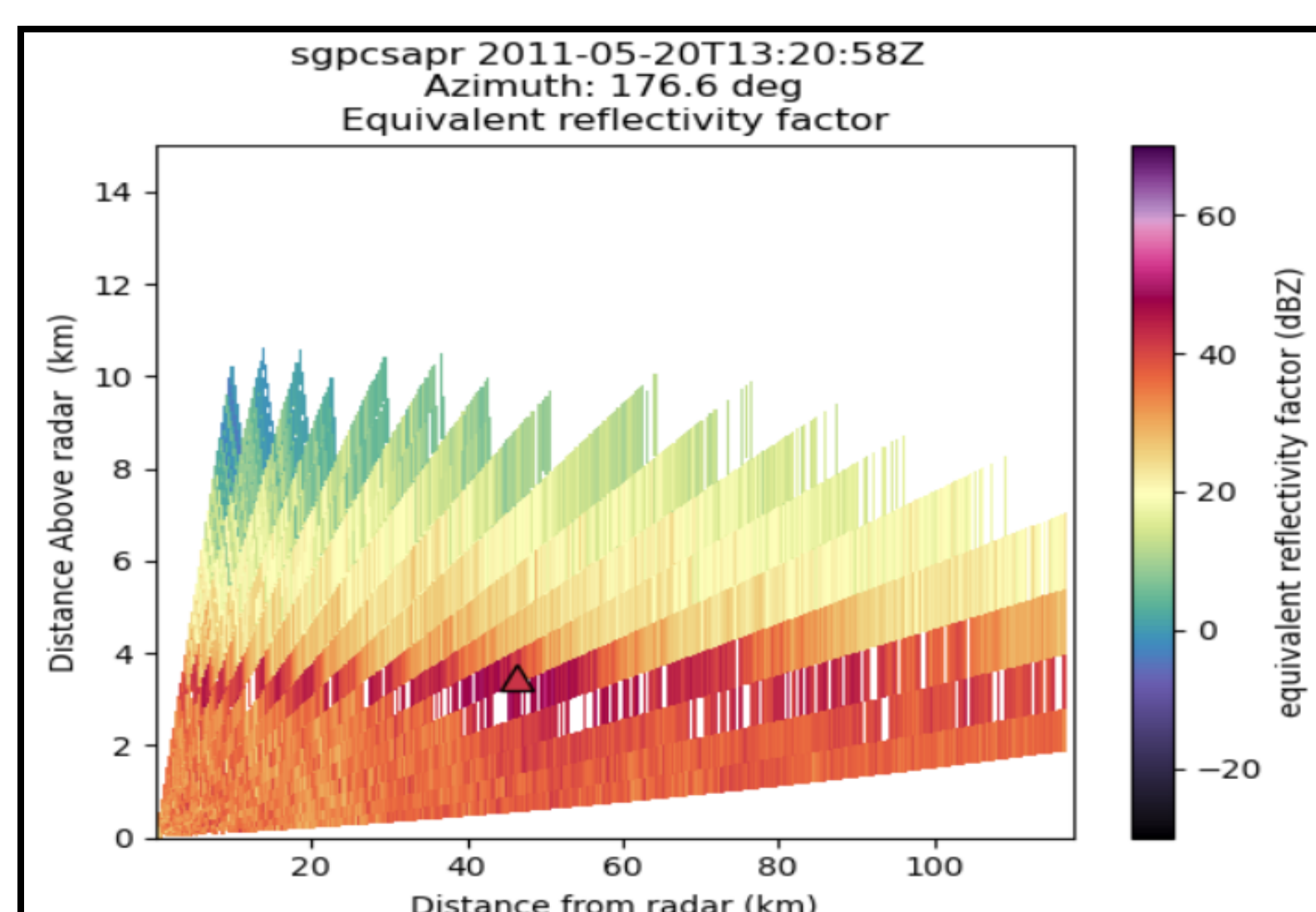


Figure 2: C-SAPR pseudo RHI radar scan with aircraft location denoted by marker.

FINDINGS

- There were three temporal regions where CSU radar tools detected “hail” 13:19-13:40 CDT, 15:18-15:20 CDT, and 15:39-15:46 CDT.
- These events occurring around 3000 to 4500 meters in altitude as indicated by the UND Citation II aircraft.
- This altitude was also what appeared to be the melting layer given the high reflectivity return values at that altitude indicated by (figure 2), which shows this bright band area.
- However, HVPS-3 measurements at these “hail” locations indicated smaller graupel at lower levels and more aggregated snow as the aircraft ascended.
- Temperature seemed to have some correlation with these observations as when temperature dropped below 0°C aggregated snow was more common. Whereas graupel was more commonly observed around 3°C given the HVPS-3 images.
- At these “hail” locations, calculated reflectivity indicates values of around 22 to -20 dBZ, whereas radar reflectivity was around 35 to 50 dBZ. This discrepancy still existed even after apply a mass dimensional relationship calculation to the calculated reflectivity from the HVPS-3.

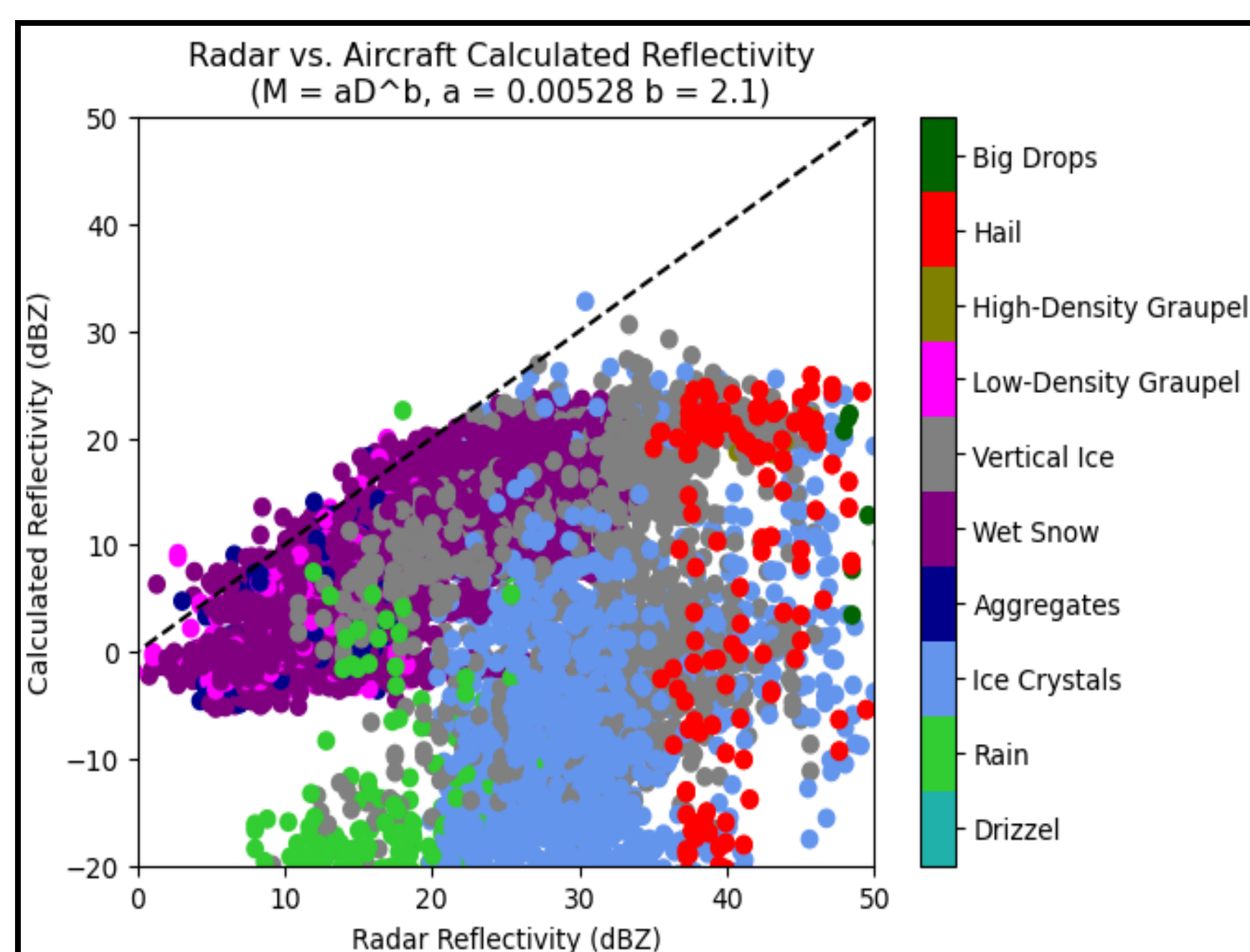


Figure 4: Radar reflectivity and aircraft calculated reflectivity comparison with hydrometeor phase classification.

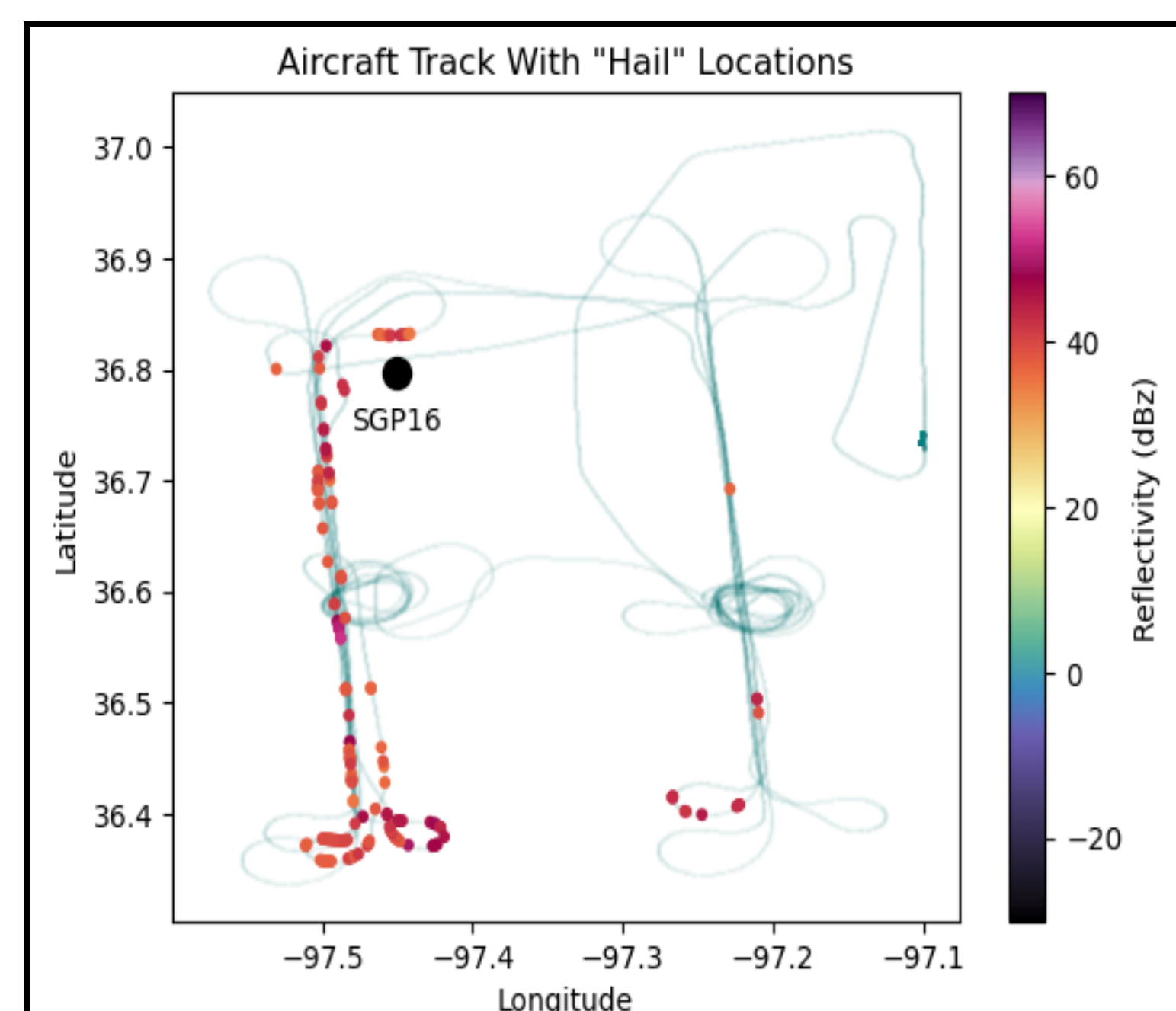


Figure 5: Aircraft track with overlaid “hail” locations and associated radar reflectivity values.

METHODS

- The Department of Energy's (DOE) Atmospheric Radiation Measurement (ARM) Southern Great Plains (SGP) site hosted the Mid-latitude Continental Convective Clouds Experiment (MC3E) from April 22 to June 6, 2011. RadCAT was built on the May 20th “golden day” event, when a squall line occurred over the site and provided a quality sampled dataset.
- The instruments used from MC3E are the C-band ARM scanning precipitation radar (C-SAPR) and the University of North Dakota Citation II microphysics aircraft equipped with the High-Volume Precipitation Spectrometer (HVPS-3).
- RadCAT was developed to be an extension of the Python-ARM Radar Toolkit (Py-ART), which creates a radar object containing CSAPR data ready for extraction.
- Vertical column profile datasets then get extracted using the Py-ART `columnsect.get_field_location` function by locating the nearest radar gates to the location of the aircraft at a given valid time for that radar scan.
- Three radar sweeps around the aircraft are extracted along with the hydrometeor classification of the gate that the aircraft is in. The hydrometeor classification algorithm is provided by CSU radar tools where a fuzzy logic algorithm is utilized.

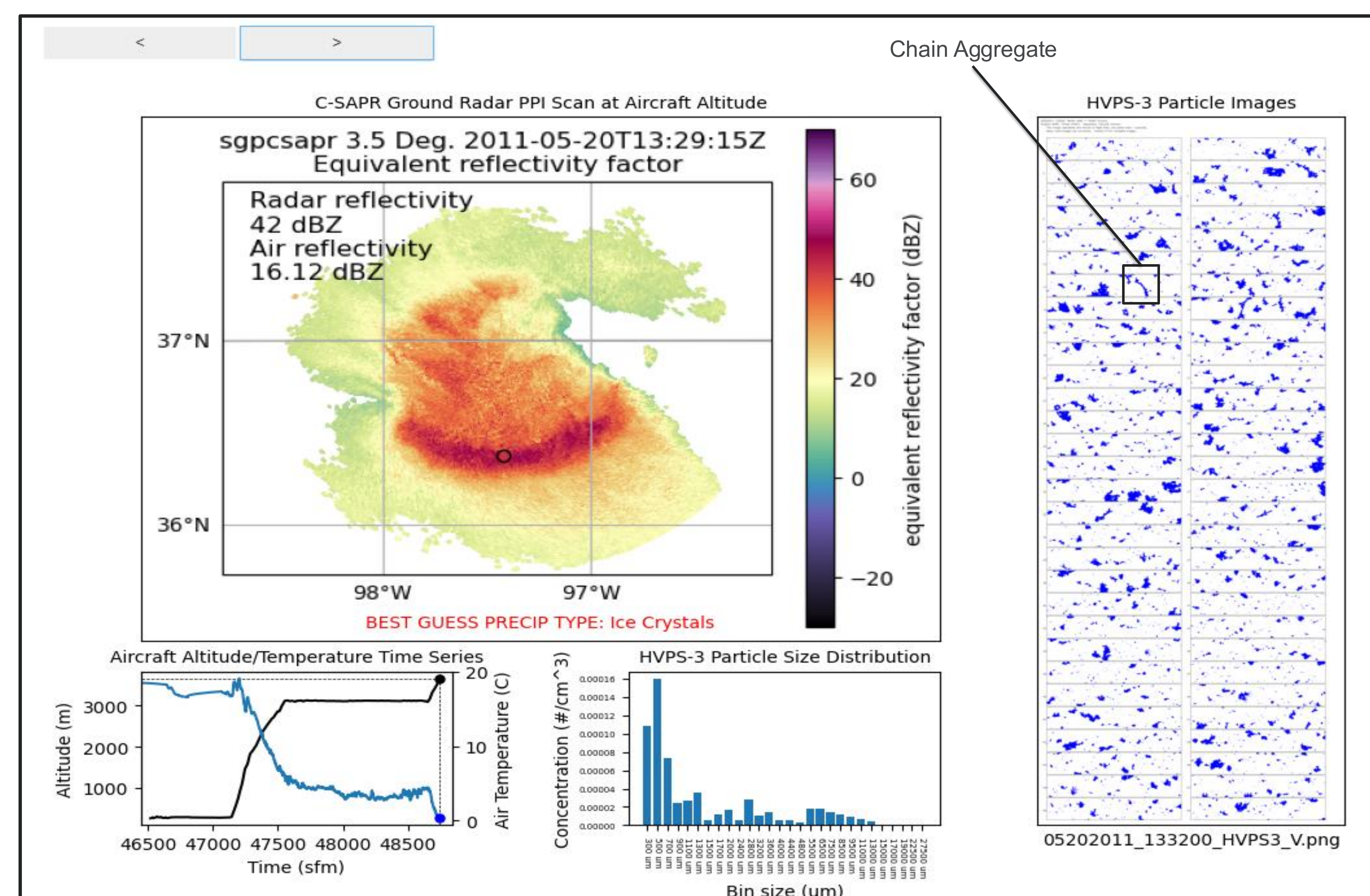


Figure 3: Interactive data display panel showing aircraft altitude and temperature time series, HVPS-3 particle size distribution and images, and C-SAPR radar PPI scan.

CONCLUSIONS / NEXT STEPS

- CSU radar tools has some periods of false classification of hail given the synced observations from the HVPS-3 particle images.
- Calculated reflectivity seems to trend lower than radar reflectivity, which could be caused by the sizing methodology of the HVPS-3 or radar overestimating liquid coated hydrometeors.
- Work on comparing more hydrometeor types between CSU radar tools and HVPS-3 observations.
- Create a configuration file that contains the mass dimensional coefficients for each CSU Radar Tool hydrometeor classification.
- Create larger datasets over many field campaigns and use machine learning models to find trends within the datasets that lead to improved hydrometeor classification.

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ABSTRACT

The Mid-latitude Continental Convective Clouds Experiment (MC3E) was a campaign hosted by the Department of Energy's (DOE) Atmospheric Radiation Measurement (ARM) Southern Great Plains (SGP) facility from April 22 – June 6, 2011. The 20 May 2011 MC3E day, in particular, has had significant attention from the scientific community to test agreements between observed radar reflectivity's and in-situ aircraft observations of hydrometeor characteristics. However, the methods used for these comparisons are not openly available for the larger scientific community. Therefore, the Radar Columns and Aircraft Time-series (RadCAT) product was developed as an open-source method for spatiotemporally syncing radar and in-situ aircraft observations. Utilizing the Python ARM Radar Toolkit (Py-ART) and CSU Radar Tools, validation of radar defined hydrometeor classification with RadCAT is conducted specifically with “hail” identified events, with particle images from the High-Volume Precipitation Spectrometer (HVPS-3) indicating these regions are a collection of small graupel and melting large aggregates.

